

Cervical Shortening 06. Twins and Multiples

Companion reading handout for simultaneous listening.

Audio course on cervical shortening during pregnancy. Episode six. Twins and multiples.

Opening frame

Twins create one of the most important traps in cervical shortening.

The cervix predicts preterm birth in twins. Sometimes it predicts very strongly. But prediction is not the same as prevention.

That is the dominant model for this episode.

In twins, cervical length is a strong risk marker, but the interventions that work in singleton pregnancy do not transfer cleanly.

This is why residents get confused. They learn that short cervix in a singleton can be treated with vaginal progesterone. They learn that cerclage can help selected singleton patients with prior spontaneous preterm birth and short cervix. They learn that pessary seems mechanically plausible. Then they see a twin pregnancy with cervical length twenty millimeters and want to apply the same logic.

But twins are not just two singleton pregnancies sharing a uterus.

The biology is different. The baseline preterm birth risk is much higher. Uterine overdistension is more prominent. Placental mass is different. The inflammatory and stretch pathways may differ. The competing risks are different. And the trials do not give the same clean answer.

Evidence and mechanisms

Let us start with prediction.

Gabbe summarizes strong risk associations. In twins, a TV cervical length of twenty millimeters or less at twenty to twenty four weeks markedly increases the probability of early preterm birth. In the cited twin meta-analysis, TV cervical length twenty millimeters or less increased the probability of birth before thirty two weeks from about six point eight percent to forty two point four percent, and birth before thirty four weeks from about fifteen point three percent to sixty one point nine percent.

A cervical length twenty five millimeters or less also increases risk. Delivery before thirty two weeks increased from about three point five percent to twenty five point eight percent, and delivery before thirty seven weeks from about forty one point two percent to seventy five point five percent.

So do not undercall the risk. A short cervix in twins matters.

But now comes the hard part.

SMFM Consult number seventy recommends against routine use of progesterone, pessary, or cerclage for treatment of cervical shortening in twin gestations outside the context of a clinical trial.

That sentence should make the resident slow down.

Why would a strong predictor not lead to a strong treatment?

Because the treatment must change outcomes, not just make mechanistic sense.

Vaginal progesterone is the cleanest intervention for asymptomatic singleton short cervix. But twin data are less consistent. Gabbe and some local guidance describe possible neonatal benefit in twins with very short cervix, such as TV cervical length twenty five millimeters or less, but they also acknowledge insufficient evidence for a specific dose or formulation and no role for unselected multiple gestations. SMFM is more conservative and recommends against routine use outside trials.

This is not a contradiction to hide. It is a guideline tension to teach.

The resident should say: cervical length predicts risk in twins, but current SMFM guidance does not support routine progesterone, cerclage, or pessary solely for twin cervical shortening outside a trial. Local MFM practice may individualize care, especially in very high-risk phenotypes, but the evidence is not equivalent to singleton short cervix.

Cerclage

Now cerclage.

Routine prophylactic or ultrasound-indicated cerclage in twins has not shown the same benefit as in selected singletons and may be harmful in some contexts. SMFM recommends against routine cerclage for twin cervical shortening outside trials.

But do not oversimplify into, never cerclage twins.

If a twin pregnancy has objective painless cervical dilation, especially with exposed membranes, the question may shift from ultrasound-indicated cerclage to physical-exam-indicated cerclage. That is still individualized and high risk, but it is not the same as placing cerclage for a closed twenty two millimeter cervix.

History also matters. A patient with a strong history of classic cervical insufficiency may require individualized MFM discussion. But the resident must not pretend that a routine twin short cervix is the same as the classic singleton prior preterm birth cerclage phenotype.

Cervical pessary

Now pessary.

Pessary has had conflicting results in singleton studies and in multiple gestation studies. ProTWIN did not show overall benefit in multiple gestations, though a subgroup signal appeared in patients below the twenty fifth percentile cervical length. Subgroup signals are useful for research questions. They are not enough for routine universal practice.

SMFM recommends against routine pessary for prevention in twin short cervix outside trials.

Bed rest and activity restriction

Now bed rest and hospitalization.

This is especially important because twin pregnancies often make clinicians anxious. The temptation is to admit the patient, reduce movement, and hope that less pressure means more time.

SMFM Consult number fifty recommends against routine inpatient hospitalization and activity restriction for prevention of preterm birth in multiple gestations. Gabbe's multiple gestation chapter also states that routine hospitalized bed rest did not decrease preterm birth in multifetal pregnancies and does not recommend cessation of work or rest at home for asymptomatic twin pregnancies with reassuring cervical length and no prior preterm birth.

So the resident reflex is this: do not prescribe bed rest because twins feel high risk. High risk is not the same as effective intervention.

Now how should twins be managed when the cervix is short?

First, confirm the measurement. It should be transvaginal and technically adequate.

Second, decide whether the patient is symptomatic. Contractions, pressure, bleeding, leakage, fever, or pain change the pathway.

Third, evaluate whether there is dilation. A short closed cervix is not the same as painless dilation with exposed membranes.

Fourth, use the finding for risk stratification, counseling, and planning. A very short cervix in twins may justify closer surveillance, MFM involvement, steroid planning if

the gestational age and delivery risk become appropriate, transfer planning if needed, and careful education about symptoms.

Fifth, avoid unsupported routine interventions: no routine progesterone, no routine cerclage, no routine pessary, no 17-OHPC, and no routine bed rest solely because the cervix is short.

Now a case.

Clinical cases

Twenty two weeks. Dichorionic diamniotic twins. TV cervical length twenty two millimeters. No contractions. No bleeding. No leakage. No dilation.

The wrong answer is to copy the singleton algorithm and prescribe vaginal progesterone as if the evidence were identical.

The better answer is to recognize high preterm birth risk, discuss the evidence uncertainty, involve MFM or local high-risk protocol, and avoid routine cerclage, pessary, 17-OHPC, and bed rest. Surveillance and counseling matter.

Second case.

Twenty three weeks. Twins. Cervical length nine millimeters. Pelvic pressure. No obvious leakage.

This is higher risk. The resident should not respond with bed rest alone. The key question is whether there is dilation, PPRM, infection, bleeding, or preterm labor. If the cervix is dilated, the problem changes. If delivery risk is high enough and gestational age fits, antenatal corticosteroids and transfer planning may enter the discussion.

Third case.

Twenty four weeks. Twins. Contractions every five minutes. Short cervix.

Now the cervical length is part of a symptomatic preterm labor triage problem. The management is not progesterone. It is assessment of true labor, membranes, infection, fetal status, steroid timing, tocolysis for a steroid or transfer window if appropriate, magnesium for neuroprotection when indicated, and neonatal counseling.

Patient counseling

Now patient counseling.

Try this:

The short cervix tells us this twin pregnancy has a higher risk of early birth. The difficult part is that treatments that help in some singleton pregnancies have not shown the same clear benefit in twins. Current SMFM guidance recommends against routine progesterone, cerclage, or pessary just because the cervix is short in twins, outside a clinical trial. We will not ignore the risk. We will use it to guide surveillance, symptom education, and planning. But we should not add treatments that have not clearly improved outcomes in this situation.

This is honest. It validates the finding. It avoids unsupported intervention.

Now let us close with the central contrast.

In singleton short cervix, prediction and prevention sometimes line up. A short cervix can identify a group where vaginal progesterone helps, and in selected prior preterm birth patients where cerclage may help.

In twins, prediction is stronger than prevention. The cervix tells you risk. It does not automatically tell you a proven treatment.

The resident reflex is this: in twins, respect the short cervix, but do not copy the singleton algorithm.